MIL-R-48401A (AR)
15 August 1977
Superseding
MIL-R-48401 (MU)
8 May 1973

MILITARY SPECIFICATION

# RING, OBTURATING FOR 60MM MORTAR

This specification is approved for use by the U. S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This is the applicable specification for split, acetal plastic obturating bands (molded) for 60mm M720, HE, mortar ammunition.

## 2. APPLICABLE DOCUMENTS

2.1 Specification and standards. The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein:

Military Standards

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-1167 | - Ammunition Data Card |
| MIL-STD-1168 | - Lot Numbering of Ammunition |
| MIL-STD-1169 | - Packaging, Packing and Marking for Shipment of Inert Ammunition Components |
| MIL-STD-1235 | - Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes |
| MIL-A-48078 | - Ammunition Standard Quality Assurance Provisions, General Specification for |

Drawings

| | |
|---|---|
| D11751152 | - Ring, Obturating for 60mm Mortar |

**Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.**

FSC: 1310

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y. 10018.)

3. REQUIREMENTS

3.1 General.-All parts shall comply with the requirements of the applicable drawings, references, specifications and the following.

3.2 Material.-The material shall be as specified on the applicable drawing. No regrind or scrap will be permitted.

**3.2.1 Certification.**-Certification must be furnished by the molder for each lot of bands certifying that the plastic used in molding complies with the requirements of 3.2.

3.3 Molded parts.-Parts made in accordance with the applicable drawings shall be produced by the injection molding process. All tests required by this document shall be performed after the parts have been maintained at 73°F ± 5°F, for a minimum of 24 hours. The tests shall be performed at 73°F ± 5°F.

3.3.1 Workmanship.-The molded parts shall be uniform in color, free of cracks, burn marks, flashing and contamination. Surface irregularities resulting from cutting off sprues and gates shall be removed by grinding and buffing.

3.4 Hidden defects.-When parts which have been inspected, tested and accepted contain inherent defects subsequently exposed, the contractor may be required to replace the defective material at no expense to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible



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for the performance of all inspection requirements specified herein. Except as otherwise specified, in the contract or order, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definition.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First Article sample

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 100 obturating rings. The sample shall represent all mold cavities used in production. Each obturating ring shall be identified as to the mold cavity producing it. All parts and materials including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply. A first article or portion thereof as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in the manufacturing processes, procedures, material used, drawings, specifications or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.3 Inspection

4.3.1 Lot

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235 as applicable.

4.3.1.2 Band lot.-A lot shall consist of bands produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Each lot shall contain bands manufactured from one batch or "run" of resin and molded in one mold on one machine. Each obturating ring shall be identified as to the mold cavity producing it. Each lot shall be identified with an ammunition lot number per MIL-STD-1168.



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4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activities. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for roughness determination.

4.3.2.1 Tool control dimensions.-Dimensions marked tool control shall be gaged during first article testing and after each 1000 items produced from each cavity mold for each machine in operation and after any tooling change. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found."

4.3.2.2 Ring, Obturating for 60mm Mortar, Drawing D11751152

| Defect | | Method of Inspection |
|---|---|---|
| CRITICAL | | |
| None defined | | |
| MAJOR | | |
| 101. | Thickness | Gage |
| 102. | Width | Gage |
| 103. | Inside diameter with outside diameter, (functional) | Gage |
| 104. | Radius of corners on outside diameter, max. | Gage |
| MINOR | | |
| 201. | Overlap | Visual |

4.3.3 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Defect/Inspection Method' column starting with paragraph 4.3.2.2. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.



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5. PREPARATION FOR DELIVERY

5.1 Preservation and packing, Level C. Preservation and packaging of the bands shall be in accordance with the requirements of MIL-STD-1169.

In addition

a. Bands shall be placed on cardboard tubes, the dimensions of which are as follows:

Length - 16 inches nominal
Thickness - 0.10 inches nominal
Outer Diameter - 1.875 ± .015

The tubes shall be smooth with no delaminations, breaks or bends on the outside surface.

b. The bands shall be placed on the tubes such that no overlapping of bands shall occur.

c. Ninety five to one hundred and five bands shall be placed on each tube and the tubes with bands shall be placed into individual polyethylene bags and sealed.

d. The tubes shall then be placed into the compartments of a corrugated cardboard box measuring 16 x 16 x 16 inches high which has been partitioned with corrugated cardboard into 49 uniform compartments.

e. The boxes will then be sealed in the standard manner ready for shipment.

5.2 Packing, Level C.-Packing of the bands shall be in accordance with MIL-STD-1169.

5.3 Marking of shipments.-Shipping containers shall be marked in accordance with MIL-STD-1169. In addition the ammunition lot number shall be identified on the shipping container (see 4.3.1.2).

5.4 Data Card.-Data Card information shall be in accordance with MIL-STD-1167.

6. NOTES



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6.1 Visual examination qualifications.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.3 Submission of Designs for Approval.-See 6.2.3 of MIL-A-48078. Submit equipment designs, as required to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801

Custodian:

Army - AR

Preparing Activity:

Army - AR

Project No. 1310-A110



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

**INSTRUCTIONS:** The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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